AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM351
Data Management and Analysis

Mid-Term Examination (MTA) – Makeup
Fall Semester 2025/2026

Date: 4/Dec./2025

umber of Exam Pages: 4 ncluding this cover sheet)	**Time Allowed: 2 Hours**

Instructions:

- Total Marks: **50**
- This exam consists of o 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are NOT allowed.

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Part I: [5 marks]

Please select the most appropriate answer and write it on the answer booklet

1. The four stages of the data pipeline are: Acquisition, Preparation, Analysis and _____.

a. Archiving b. Data management c. Presentation d. Repurposing

2. The _____ character encoding encodes all 1,112,064 valid code points in Unicode using a vari-able number of bytes (from 1-4) and is backward compatible with ASCII code.

a. Unicode b. ASCII c. UTF-8 d. Huffman code

3. Which are the key issues in Data Management?

a. Legal b. Control c. Curation d. All of the above

4. In the following data set: {10, 109, 10, 108, 107,15, 251, 13} the mode is: _____.

a. 107 b. 10 c. 108 d. 251

5. Spreadsheets are easily shared, possibly outside the business for which they were created which creates a problem of data _____.

a. cleansing b. sharing c. control d. acquisition

6. JSON File stands for _____.

a. JavaSript Operator Notation b. JavaScript Object Notation c. JavaServer Object Notation d. None of the above

7. In the relational model, we cannot assign NULL to an attribute that is _____.

a. part of the primary key b. a foreign key c. composite d. a non-key attribute

8. Query languages, such as SQL, which specify what data to be retrieved but not how it is to be re-trieved are referred to as: _____ languages.

a. procedural (prescrip-tive) b. non-procedural (descriptive c. host d. None of the above

9. A relation is in First Normal Form (1NF) if each attribute contains only _____ values, that is, it has no repeating groups of values.

a. Atomic b. Numerical c. String d. Unique

10. _____ is a referential action that specifies that in the case a referenced tuple is deleted, all referenced tuples are deleted as well

a. Cascade b. Restrict c. Null d. Default

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Part Two: [35 marks]

Question One: [10 marks]

What will be the output of the following snippets of python code? Write down the exact output, including all brackets, parentheses, braces, quotes and commas ([], (), {}, '', "", ,) . [1 mark for each correct output line]:

1.	x= ['RAM', 'ROM', 'CPU'] print (x) x.append ('Mouse') print (x) x = x + ['Monitor', 'Keyboard'] print (x)	2.	tuple1 = ('GR101', 'M359', 'TM351') a, b, c = tuple1 print(tuple1) print(c) print (a)
3.	x = 2 while x < 9: print (x) x+=3	4.	x=[11, 22, 33, 44, 11] print (x[0], x[-1]) print (x[1], x[-2]) print (x[0] == x[-1])
5.	set1 = {'TM351', 'M359', 'T471'} set2 = {'MT262', 'TM351'} print (set1-set2) print (set1 & set2) print (set1.issuperset(set2))		

Question Two: [15 Marks]

Answer the following questions related to the purchase of art in a hotel as shown below:

ART_PURCHASE (customer_id, art_id, art_theme, art_price, artist_code, artist_name, customer_name, customer_address, customer_phone, purchase_date)

Suppose that the following functional dependencies hold:

- **customer_id** determines **customer_name, customer_address,** and **customer_phone**
- **art_id** determines **art_theme, art_price, artist_code** and **artist_name**
- **artist_code** determines **artist_name**
- **customer_id** and **art_id** determines **purchase_date**

a. What is the highest normal form that this relation conforms to and why? [3 Marks]
b. Normalise this relation to 3NF, showing the new relations in the manner the relation *ART_PURCHASE* is shown above. Make sure you identify the primary key for each normalized relation by underlining it [6 Marks]
c. Write simple select SQL queries to produce the required normalized relations [6 marks]

Question Three: [10 Marks]

Consider the CLUB database schema below.

CLUB (club_id, club_name, location, club_type)

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Each row records the following data about a club identified by club_id which is the primary key of the relation.

Column	Description
club_id (Primary key)	Identifier of the club
club_name	Name of the club
location	Location of the club
club_type	Type of the club

MEMBERSHIP (club_id, member_id, date_of_join)
A club contains one or more members. Each row records the date of join of a member in a club, which is identified by club_id and member_id.

Column	Description
club_id (Primary key)	Id of the club
member_id (Primary key)	Id of the member
date_of_join	Date when a member joined a club

MEMBER (member_id, member_name, member_addr, member_phone)
Each row records the following data about a member identified by member_id which is the primary key of the relation.

Column	Description
member_id (Primary key)	Id of the member
member_name	Name of the member
member_addr	Address of the member
member_phone	Phone of the member

Use the above CLUB schema and write SQL queries for the following:

1. List club id's and club names of all clubs in descending order of club name.
2. List member id's, member name's and date of join of all members who joined on 19 February 2020.
3. Change the phone of the member with id 20855 to 99663316.
4. Remove the clubs whose type is "C".
5. Count the number of clubs based on club_type and display the club_type and the corresponding count.

Part Three [10 marks]

Q1. List and explain Approaches to handling dirty data. (5 marks)

Q2. List any five Jacques Bertin described eight visual dimensions that apply to two-dimensional representations. (5 marks)

End of questions

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